

```
In [81]: # importing lib,
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

In [82]: df = pd.read_csv('mymovieDb.csv', lineterminator='\n')
df.head()

Out[82]:
```

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	Genre	Poster_Url
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	Action, Adventure, Science Fiction	https://image.tmbd.org/t/p/original/1g0dY1tq4i...
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	Crime, Mystery, Thriller	https://image.tmbd.org/t/p/original/74xTEg7R3...
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	Thriller	https://image.tmbd.org/t/p/original/VDHsLnOWK1...
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	Animation, Comedy, Family, Fantasy	https://image.tmbd.org/t/p/original/4QDPNkMf5...
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	Action, Adventure, Thriller, War	https://image.tmbd.org/t/p/original/aq4Pwv5Xeu...

```
In [83]: # viewing dataset info
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Release_Date          9827 non-null   object
 1   Title                 9827 non-null   object
 2   Overview              9827 non-null   object
 3   Popularity            9827 non-null   float64
 4   Vote_Count            9827 non-null   int64
 5   Vote_Average          9827 non-null   float64
 6   Original_Language     9827 non-null   object
 7   Genre                 9827 non-null   object
 8   Poster_Url           9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB

In [84]: # exploring genres column
df['Genre'].head()

Out[84]:
0    Action, Adventure, Science Fiction
1           Crime, Mystery, Thriller
2                      Thriller
3    Animation, Comedy, Family, Fantasy
4    Action, Adventure, Thriller, War
Name: Genre, dtype: object

In [85]: # check for duplicated rows
df.duplicated().sum()

Out[85]: np.int64(0)

In [86]: # exploring summary statistics
df.describe()

Out[86]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805636	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

```
In [87]: df.head()

Out[87]:
```

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	Genre	Poster_Url
0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	Action, Adventure, Science Fiction	https://image.tmbd.org/t/p/original/1g0dY1tq4i...
1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	Crime, Mystery, Thriller	https://image.tmbd.org/t/p/orignal/74xTEg7R3...
2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	Thriller	https://image.tmbd.org/t/p/original/VDHsLnOWK1...
3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	Animation, Comedy, Family, Fantasy	https://image.tmbd.org/t/p/orignal/4QDPNkMf5...
4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	Action, Adventure, Thriller, War	https://image.tmbd.org/t/p/orignal/aq4Pwv5Xeu...

```
In [88]: # casting column a
df['Release_Date'] = pd.to_datetime(df['Release_Date'])
# confirming changes
print(df['Release_Date'].dtypes)
datetime64[ns]

In [89]: df['Release_Date'] = df['Release_Date'].dt.year
df['Release_Date'].dtypes

Out[89]: dtype('int32')

In [90]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Release_Date          9827 non-null   int32
 1   Title                 9827 non-null   object
 2   Overview              9827 non-null   object
 3   Popularity            9827 non-null   float64
 4   Vote_Count            9827 non-null   int64
 5   Vote_Average          9827 non-null   float64
 6   Original_Language     9827 non-null   object
 7   Genre                 9827 non-null   object
 8   Poster_Url           9827 non-null   object
dtypes: float64(2), int32(1), int64(1), object(5)
memory usage: 652.7+ KB

In [91]: df.head()

Out[91]:
```

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Language	Genre	Poster_Url
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	en	Action, Adventure, Science Fiction	https://image.tmbd.org/t/p/original/1g0dY1tq4i...
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	en	Crime, Mystery, Thriller	https://image.tmbd.org/t/p/original/74xTEg7R3...
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	en	Thriller	https://image.tmbd.org/t/p/original/VDHsLnOWK1...
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	en	Animation, Comedy, Family, Fantasy	https://image.tmbd.org/t/p/original/4QDPNkMf5...
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	en	Action, Adventure, Thriller, War	https://image.tmbd.org/t/p/orignal/aq4Pwv5Xeu...

```
In [92]: # making list of column to be dropped
cols = ['Overview', 'Original_Language', 'Poster_Url']
# dropping columns and confirming changes
df.drop(cols, axis = 1, inplace = True)
df.columns

Out[92]: Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
        'Genre'],
        dtype='object')

In [93]: df.head()

Out[93]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	6.3	Thriller
3	2021	Encanto	2402.201	5076	7.7	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	7.0	Action, Adventure, Thriller, War

```
In [94]: def categorise_col (df, col, labels):
edges = [df[col].describe()['min'],
          df[col].describe()['25%'],
          df[col].describe()['50%'],
          df[col].describe()['75%'],
          df[col].describe()['max']]
df[col] = pd.cut(df[col], edges, labels = labels, duplicates='drop')
return df

In [95]: # define labels for edges
labels = ['not_popular', 'below_avg', 'average', 'popular']
# categorize column based on labels and edges
categorise_col(df, 'Vote_Average', labels)
# confirming changes
df['Vote_Average'].unique()

Out[95]: ['popular', 'below_avg', 'average', 'not_popular', NaN]
Categories (4, object): ['not_popular' < 'below_avg' < 'average' < 'popular']

In [96]: df.head()

Out[96]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_avg	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

```
In [97]: # exploring column
df['Vote_Average'].value_counts()

Out[97]: Vote_Average
not_popular    2467
popular         2450
average         2412
below_avg       2398
Name: count, dtype: int64

In [98]: # dropping NaNs
df.dropna(inplace = True)
# confirming
df.info().sum()

Out[98]: Release_Date    0
         Title           0
         Popularity      0
         Vote_Count      0
         Vote_Average     0
         Genre           0
dtype: int64

In [99]: # split the strings into lists
df['Genre'] = df['Genre'].str.split(',')
# explode the lists
df = df.explode('Genre').reset_index(drop=True)
df.head()

Out[99]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

```
In [100]: # casting column into category
df['Genre'] = df['Genre'].astype('category')
# confirming changes
df['Genre'].dtypes

Out[100]: CategoricalType(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
        'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
        'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
        'TV Movie', 'Thriller', 'War', 'Western'],
        ordered=False, categories_dtype=object)

In [101]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Release_Date          25552 non-null   int32
 1   Title                 25552 non-null   object
 2   Popularity            25552 non-null   float64
 3   Vote_Count            25552 non-null   int64
 4   Vote_Average          25552 non-null   category
 5   Genre                 25552 non-null   category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB

In [102]: df.nunique()

Out[102]: Release_Date    100
         Title           9415
         Popularity      9098
         Vote_Count      3265
         Vote_Average      4
         Genre           19
dtype: int64
```

Data visualization

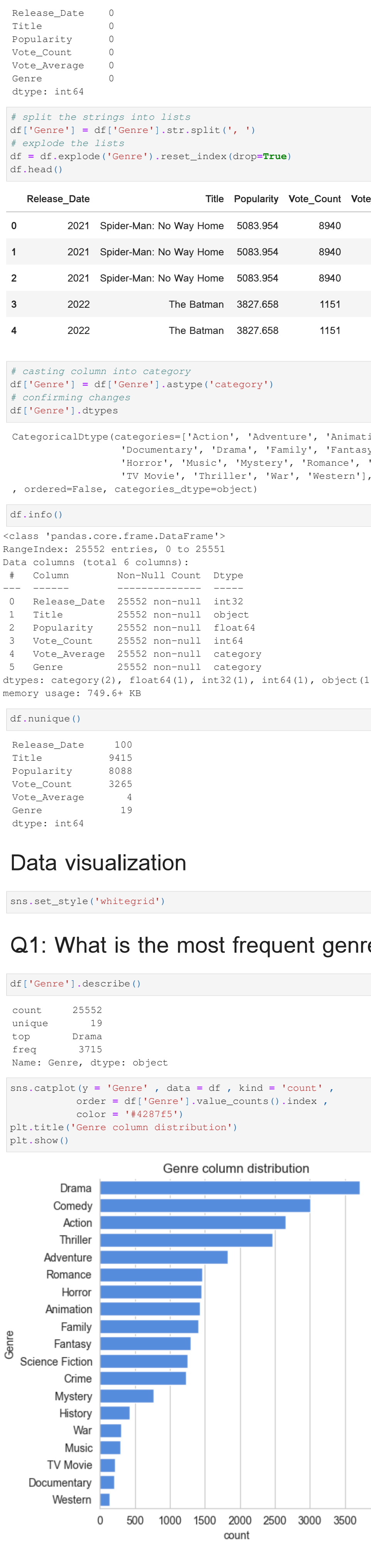
```
In [103]: sns.set_style('whitegrid')

Q1: What is the most frequent genre in the dataset?
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```
In [104]: df['Genre'].describe()

Out[104]: count      25552
         unique        19
         top      Drama
         freq       3715
         Name: Genre, dtype: object

In [105]: sns.catplot(y = 'Genre', data = df, kind = 'count',
         order = df['Genre'].value_counts().index,
         color = '#4287f5')
plt.title('Genre column distribution')
plt.show()
```



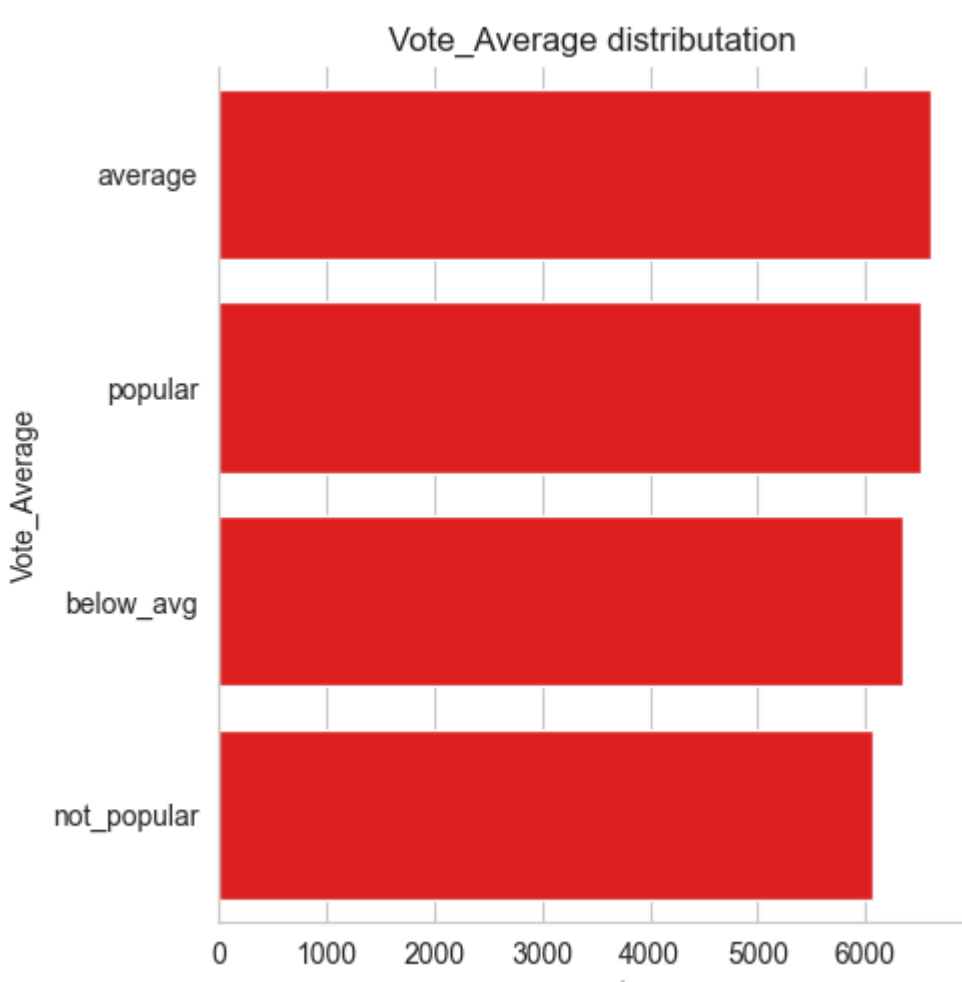
Q2: What genres has highest votes ?

```
In [106]: df.head()

Out[106]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

```
In [107]: sns.catplot(y = 'Vote_Average', data = df, kind = 'count',
         order = df['Vote_Average'].value_counts().index,
         color = 'red')
plt.title('Vote_Average distribution')
plt.show()
```



Q3: What movie got the highest popularity ? what's its genre ?

```
In [108]: df.head()

Out[108]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction
3	2022	The Batman	3827.658	1151	popular	Crime
4	2022	The Batman	3827.658	1151	popular	Mystery

```
In [109]: df[df['Popularity'] == df['Popularity'].max()]

Out[109]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action
1	2021	Spider-Man: No Way Home	5083.954	8940	popular	Adventure
2	2021	Spider-Man: No Way Home	5083.954	8940	popular	Science Fiction

Q4: What movie got the lowest popularity? what's its genre?

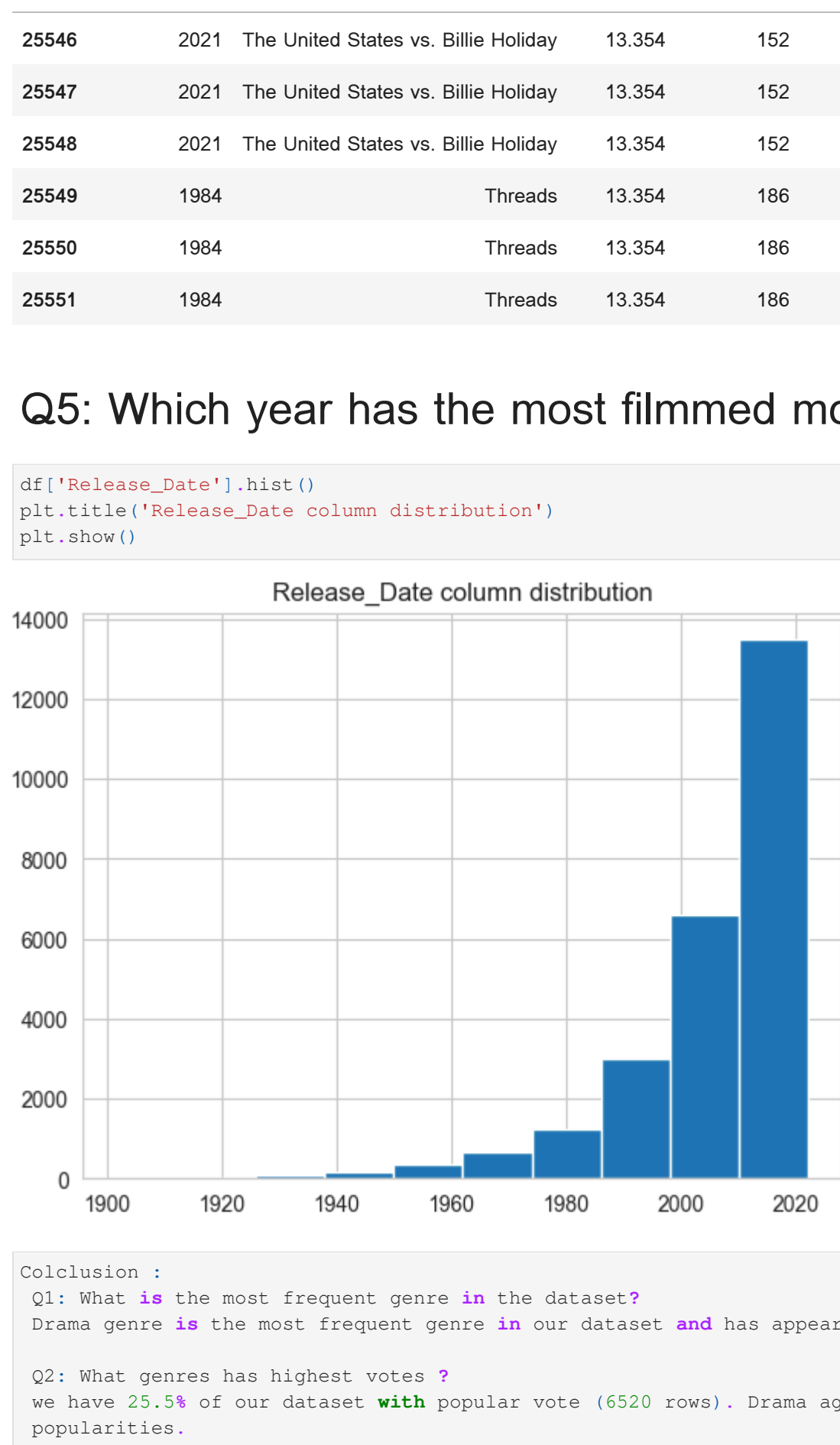
```
In [110]: df[df['Popularity'] == df['Popularity'].min()]

Out[110]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
25546	2021	The United States vs. Billie Holiday	13.354	152	average	Music
25547	2021	The United States vs. Billie Holiday	13.354	152	average	Drama
25548	2021	The United States vs. Billie Holiday	13.354	152	average	History
25549	1984	Threads	13.354	186	popular	War
25550	1984	Threads	13.354	186	popular	Drama
25551	1984	Threads	13.354	186	popular	Science Fiction

Q5: Which year has the most filmed movies?

```
In [111]: df['Release_Date'].hist()
plt.title('Release_Date column distribution')
plt.show()
```



```
In [ ]: Conclusion :
Q1: What is the most frequent genre in the dataset?
Drama genre is the most frequent genre in our dataset and has appeared more than 14% of the times among 19 other genres.
Q2: What genres has highest votes ?
We have 15.9% of our dataset with popular vote (6520 rows). Drama again gets the highest popularity among fans by being having more than 18.5% of movies popularities.
Q3: What movie got the highest popularity ? what's its Action , genre ?
Spider-Man: No Way Home has the highest popularity rate in our dataset and it has genres of Adventure and Science Fiction .
Q3: What movie got the lowest popularity ? what's its genre ?
The united states, threads has the highest lowest rate in our dataset and it has genres of music , drama , 'war', 'sci-fi' and history'.
Q4: Which year has the most filmed movies?
year 2020 has the highest filming rate in our dataset.
```